Lesson 3.7 - The Mean Formulas

> **Video:** Arithmetic Mean, Geometric Mean, Weighted Mean, Harmonic Mean, Root Mean Square Formula by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

This lesson covers five kinds of "average", how each relates to a type of sequence, and when to use each.

Mean	Formula (n values)	Two values a, b
Arithmetic	$\frac{x_1 + \cdots + x_n}{n}$	$\frac{a+b}{2}$
Geometric	$(x_1 x_2 \cdots x_n)^{1/n}$	$\sqrt{ab}$
Weighted	$\frac{\sum wx}{\sum w}$	—
Harmonic	$\frac{n}{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}$	$\frac{2}{\frac{1}{a}+\frac{1}{b}} = \frac{2ab}{a+b}$
Root mean square	$\sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}}$	$\sqrt{\frac{a^2+b^2}{2}}$

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1. Arithmetic mean and arithmetic sequences

An **arithmetic sequence** adds the same **common difference** d each time.

Example: 2, 5, 8, 11, 14, 17, 20 ($d = 3$)

- Mean of all seven terms: $\frac{77}{7} = 11$, which is the **middle term** a_4.
- The arithmetic mean of two terms equidistant from a middle term gives that middle term:
  - $\frac{a_1 + a_3}{2} = \frac{2+8}{2} = 5 = a_2$
  - $\frac{a_1 + a_5}{2} = \frac{2+14}{2} = 8 = a_3$
  - $\frac{a_3 + a_7}{2} = \frac{8+20}{2} = 14 = a_5$

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2. Geometric mean and geometric sequences

A **geometric sequence** multiplies by the same **common ratio** r each time.

Example: 3, 6, 12, 24, 48, 96, 192 ($r = 2$)

- $\sqrt{a_1 a_3} = \sqrt{3 \cdot 12} = \sqrt{36} = 6 = a_2$
- $\sqrt{a_1 a_5} = \sqrt{3 \cdot 48} = \sqrt{144} = 12 = a_3$
- Geometric mean of all seven terms:

  $$(3 \cdot 6 \cdot 12 \cdot 24 \cdot 48 \cdot 96 \cdot 192)^{1/7} = (4{,}586{,}471{,}424)^{1/7} = 24 = a_4$$

  This is the middle term again.

With $n = 2$, $(x_1 x_2)^{1/2}$ is just $\sqrt{ab}$.

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3. Weighted mean

$$\bar{x}_w = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_n x_n}{w_1 + w_2 + \cdots + w_n}$$

Use it when the data points are **not equally important**: center of mass, average atomic mass, concentration of a mixture.

Example - mixing salt solutions. 150 mL of 20% NaCl is mixed with 50 mL of 40% NaCl. Here x is the concentration and w is the volume:

$$C = \frac{C_1 V_1 + C_2 V_2}{V_1 + V_2} = \frac{20(150) + 40(50)}{150 + 50} = \frac{3000 + 2000}{200} = 25\%$$

An equal-weight average would give 30%. There is more of the 20% solution, so the result is pulled toward 20%. With equal volumes (100 mL each), the answer would be exactly 30%.

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4. Harmonic mean and harmonic sequences

The harmonic mean is the **reciprocal of the average of the reciprocals**:

$$H = \left(\frac{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}{n} \right)^{-1} = \frac{n}{\sum \frac{1}{x}}$$

From the definition to the product form

Put the reciprocals over a common denominator.

Two values a, b (common denominator ab):

$$\frac{1}{a} + \frac{1}{b} = \frac{b + a}{ab} \quad \Rightarrow \quad H = \frac{2}{\frac{a+b}{ab}} = \frac{2ab}{a + b}$$

Three values a, b, c (common denominator abc):

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc + ac + ab}{abc} \quad \Rightarrow \quad H = \frac{3abc}{ab + bc + ca}$$

General n**:** each $\frac{1}{x_j}$ becomes the product of all the *other* values over the full product, so

$$H = \frac{n \prod_{i=1}^{n} x_i}{\sum_{j=1}^{n} \prod_{i \neq j} x_i}$$

For $n = 2$, "the product of the others" is just one number (b or a), so the denominator collapses to $a + b$.

> **Warning:** $\frac{n \prod x_i}{\sum x_i}$ is **only** correct for $n = 2$. Check with 1, 2, 4:
>
> - True: $H = \frac{3}{1 + 0.5 + 0.25} = \frac{3}{1.75} \approx 1.714$
> - Correct product form: $\frac{3(8)}{2 + 4 + 8} = \frac{24}{14} \approx 1.714$
> - Wrong shortcut: $\frac{3(8)}{1 + 2 + 4} = \frac{24}{7} \approx 3.43$, larger than the arithmetic mean (2.33), which a harmonic mean can never be.

> For $n > 2$, just use $H = \dfrac{n}{\sum \frac{1}{x}}$.

A **harmonic sequence** consists of the reciprocals of an arithmetic sequence:

$$\tfrac{1}{2}, \tfrac{1}{5}, \tfrac{1}{8}, \tfrac{1}{11}, \tfrac{1}{14}, \tfrac{1}{17}, \tfrac{1}{20}$$

(The denominators 2, 5, 8, ... form an arithmetic sequence.)

- $\dfrac{2}{\frac{1}{h_1} + \frac{1}{h_3}} = \dfrac{2}{2+8} = \dfrac{1}{5} = h_2$
- $\dfrac{2}{\frac{1}{h_2} + \frac{1}{h_6}} = \dfrac{2}{5+17} = \dfrac{1}{11} = h_4$
- All seven terms: $\dfrac{7}{2+5+8+11+14+17+20} = \dfrac{7}{77} = \dfrac{1}{11} = h_4$

Application - average speed for equal distances

A car drives **400 mi** from A to B at **80 mph**, then returns **400 mi** at **50 mph**. What is its average speed?

Physics method (total distance ÷ total time):

- Time out: $\dfrac{400}{80} = 5$ h. Time back: $\dfrac{400}{50} = 8$ h.
- $\bar{v} = \dfrac{800}{13} \approx 61.54$ mph

Weighted-mean method (weights = time spent at each speed):

$$\bar{v} = \frac{80(5) + 50(8)}{5+8} = \frac{800}{13} \approx 61.54 \text{ mph}$$

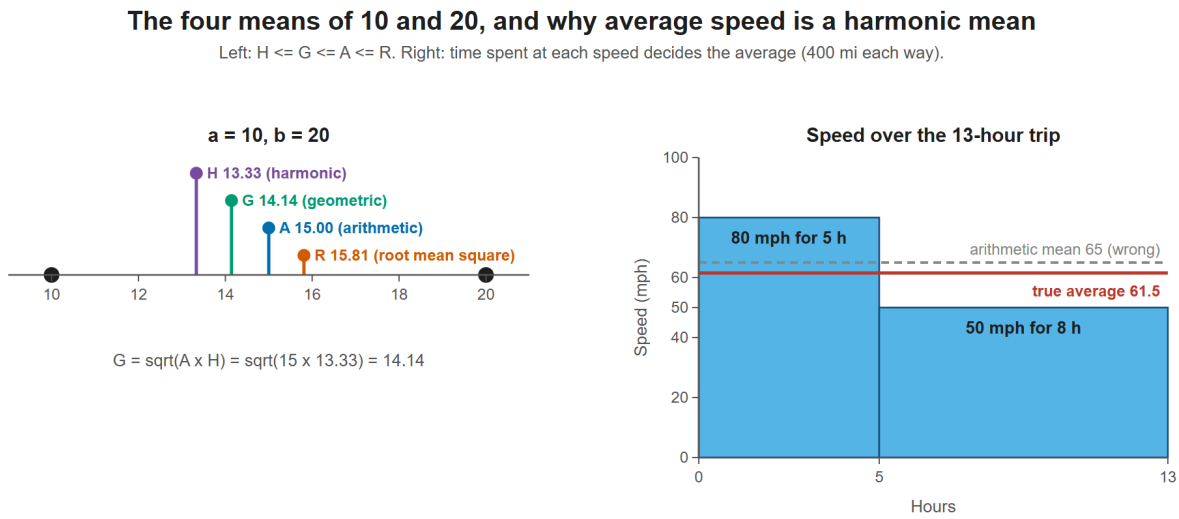
Or using time fractions: $80\left(\frac{5}{13}\right) + 50\left(\frac{8}{13}\right) = 80(0.3846) + 50(0.6154) \approx 61.54$ mph. The car spends about 38.5% of the time at 80 mph and about 61.5% at 50 mph.

Harmonic-mean method:

$$\bar{v} = \frac{2}{\frac{1}{80} + \frac{1}{50}} = \frac{2(80)(50)}{80+50} = \frac{8000}{130} = \frac{800}{13} \approx 61.54 \text{ mph}$$

The arithmetic mean (65 mph) is **wrong** here. The car spends **more time** at the slower speed, so the true average is closer to 50.

> When **equal distances** are covered at different speeds, the average speed is the **harmonic mean** of the speeds.



Left: for 10 and 20 the four means always come in the order H, G, A, R. Right: on the round trip the car spends 8 hours at 50 mph but only 5 hours at 80 mph, so the true average speed (61.5 mph, a harmonic mean) is below the plain average of 65.

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5. Root mean square (RMS)

"Root" = square root, "square" = square the values:

$$x_{\text{RMS}} = \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$$

Example. 5, 8, 12, 16, 21

$$\sqrt{\frac{25 + 64 + 144 + 256 + 441}{5}} = \sqrt{\frac{930}{5}} = \sqrt{186} \approx 13.64$$

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6. Comparing the four means

Example. $a = 10$, $b = 20$

Mean	Calculation	Value
Harmonic H	$\frac{2(10)(20)}{30} = \frac{40}{3}$	13.33
Geometric G	$\sqrt{200} = 10\sqrt{2}$	14.14
Arithmetic A	$\frac{30}{2}$	15
RMS R	$\sqrt{\frac{100 + 400}{2}} = \sqrt{250} = 5\sqrt{10}$	15.81

$$H \le G \le A \le R$$

Relationships between the three classical means:

$$G = \sqrt{A \cdot H}, \qquad H = \frac{G^2}{A}, \qquad A = \frac{G^2}{H}$$

Check: $\sqrt{15 \cdot \frac{40}{3}} = \sqrt{200} = 10\sqrt{2} = G$ ✓

If you know any two of A, G, H, you can find the third.

Key takeaways

- **Arithmetic mean:** equal weights. It gives the middle term of an arithmetic sequence.
- **Geometric mean:** multiply, then take the n-th root. It gives the middle term of a geometric sequence.
- **Weighted mean:** use when the values carry different importance (volume, time, credits).
- **Harmonic mean:** average of rates over **equal distances** (or equal amounts). It gives the middle term of a harmonic sequence.
- **RMS:** square, average, then square root. It is the largest of the four.
- $H \leq G \leq A \leq R$ and $G^2 = A \cdot H$.